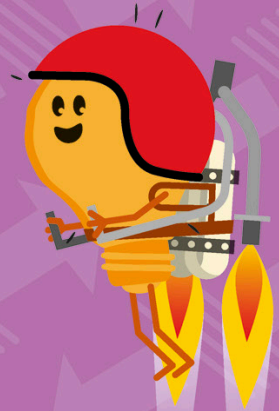
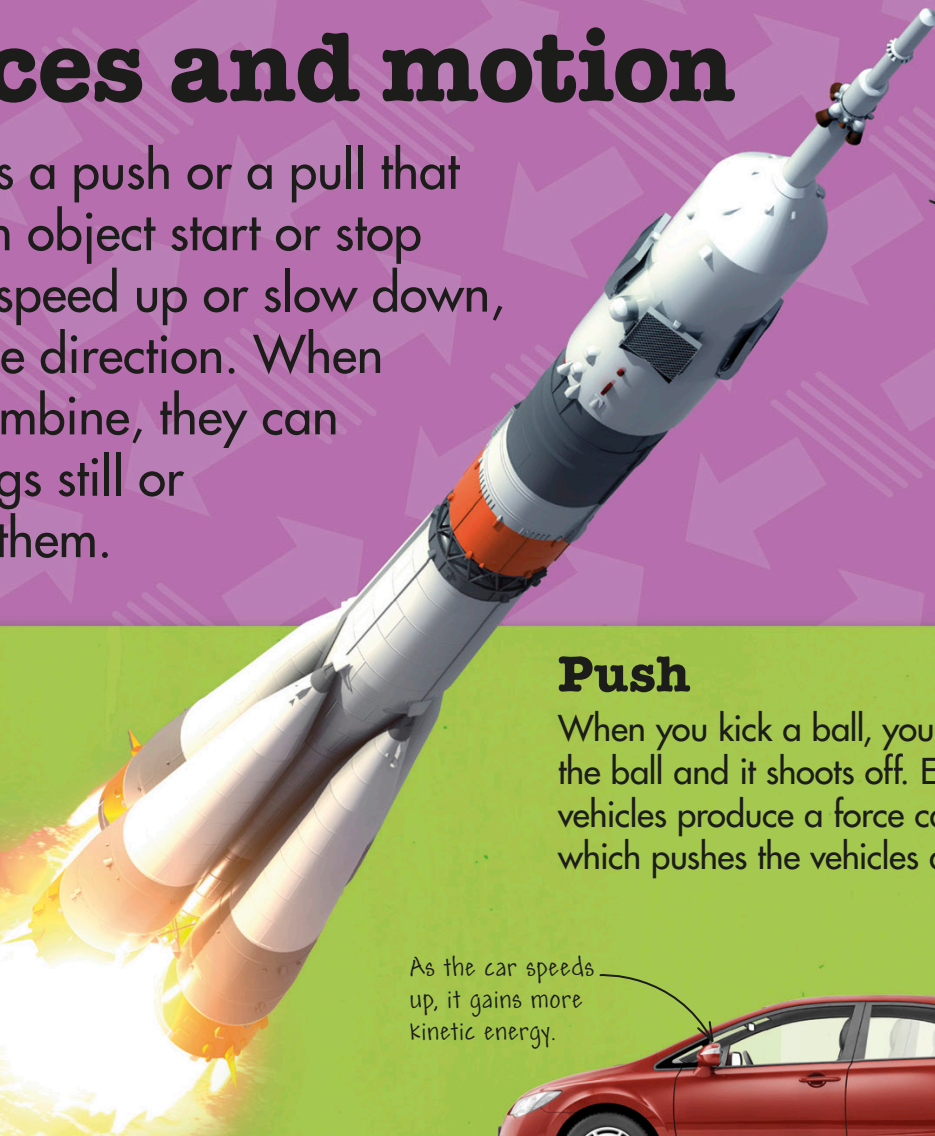


Forces and motion

A **force** is a push or a pull that makes an object start or stop moving, speed up or slow down, or change direction. When forces combine, they can hold things still or **balance** them.



This rocket's engine burns fuel to produce thrust.



Push

When you kick a ball, your foot pushes the ball and it shoots off. Engines in vehicles produce a force called **thrust**, which pushes the vehicles along.

As the car speeds up, it gains more kinetic energy.



Thrust

Rockets produce so much thrust that they can fly away from Earth and hurtle into space.

Kinetic energy

A moving object, such as this car, has a type of energy called **kinetic energy**. The faster the object is moving, the more kinetic energy it has.

Balanced forces

These tug-of-war teams are pulling with equal force in opposite directions. The forces are balanced and cancel each other out, so neither team moves.

Unless one team pulls more strongly than the other, the ribbon stays over the line.

